

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

## Device Specifications

Device Type: Lighting Control  
 Manufacturer: AAIOT  
 Firmware Version: N/A  
 Model(s): KZ0214HB1-6

## Tested on the Following Software and Firmware Versions

| IP Link Pro Control Processor Firmware | Global Scriptor Version |
|----------------------------------------|-------------------------|
| 3.01.0000-b010                         | 2.1.0                   |

## Version History

| Module Version | Date      | Notes           |
|----------------|-----------|-----------------|
| 1_0_0_0        | 1/17/2019 | Initial Version |

---

## Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.  
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.  
Example: `InterfaceName.connectionCounter = 5`

## Supported Classes and Examples

|                                                                                                               |
|---------------------------------------------------------------------------------------------------------------|
| <b>SerialClass</b>                                                                                            |
| <code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='KZ0214HB1-6')</code>               |
| <b>SerialOverEthernetClass</b>                                                                                |
| <code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='KZ0214HB1-6')</code> |

## Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

|                                                                                                                  |                                          |                                |                        |
|------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------|------------------------|
| <b>Command</b><br><b>ButtonPress</b>                                                                             | Value<br>'Long Press'<br>'Short Release' | Value<br>'Long Release'        | Value<br>'Short Press' |
| <b>Qualifier Key</b><br>'Module ID'                                                                              | Qualifier Value<br>'1' – '128'           | Qualifier Value<br>'Broadcast' |                        |
| <b>Qualifier Key</b><br>'Button'                                                                                 | Qualifier Value<br>'1' – '16'            |                                |                        |
| # ButtonPress example<br>InterfaceName.Set('ButtonPress', 'Long Press', {'Module ID': '1', 'Button': '1'})       |                                          |                                |                        |
| <b>Command</b><br><b>InidicatorControl</b>                                                                       | Value<br>'Open'                          | Value<br>'Close'               | Value<br>'Invert'      |
| <b>Qualifier Key</b><br>'Module ID'                                                                              | Qualifier Value<br>'1' – '128'           | Qualifier Value<br>'Broadcast' |                        |
| <b>Qualifier Key</b><br>'Button'                                                                                 | Qualifier Value<br>'1' – '16'            |                                |                        |
| # InidicatorControl example<br>InterfaceName.Set('InidicatorControl', 'Open', {'Module ID': '1', 'Button': '1'}) |                                          |                                |                        |
| <b>Command</b><br><b>RelayControl</b>                                                                            | Value<br>'Open'                          | Value<br>'Close'               | Value<br>'Invert'      |
| <b>Qualifier Key</b><br>'Module ID'                                                                              | Qualifier Value<br>'1' – '128'           | Qualifier Value<br>'Broadcast' |                        |
| <b>Qualifier Key</b><br>'Relay'                                                                                  | Qualifier Value<br>'1' – '4'             |                                |                        |
| # RelayControl example<br>InterfaceName.Set('RelayControl', 'Open', {'Module ID': '1', 'Relay': '1'})            |                                          |                                |                        |

## Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

|                                                                                                                                                                                                                                                                 |                 |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|
| Command                                                                                                                                                                                                                                                         | Value           | Value           |
| ConnectionStatus                                                                                                                                                                                                                                                | 'Connected'     | 'Disconnected'  |
| # ConnectionStatus examples<br>Value = InterfaceName.ReadStatus('ConnectionStatus')<br>InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)                                                                                                 |                 |                 |
| Command                                                                                                                                                                                                                                                         | Value           | Value           |
| RelayControl                                                                                                                                                                                                                                                    | 'Open'          | 'Close'         |
| Qualifier Key                                                                                                                                                                                                                                                   | Qualifier Value | Qualifier Value |
| 'Module ID'                                                                                                                                                                                                                                                     | '1' – '128'     | 'Broadcast'     |
| Qualifier Key                                                                                                                                                                                                                                                   | Qualifier Value |                 |
| 'Relay'                                                                                                                                                                                                                                                         | '1' – '4'       |                 |
| # RelayControl examples<br>InterfaceName.Update('RelayControl', {'Module ID': '1', 'Relay': '1'})<br>Value = InterfaceName.ReadStatus('RelayControl', {'Module ID': '1', 'Relay': '1'})<br>InterfaceName.SubscribeStatus('RelayControl', None, FeedbackHandler) |                 |                 |
| Command                                                                                                                                                                                                                                                         | Qualifier Value |                 |
| VersionNumber                                                                                                                                                                                                                                                   | 'String'        |                 |
| Qualifier Key                                                                                                                                                                                                                                                   | Qualifier Value | Qualifier Value |
| 'Module ID'                                                                                                                                                                                                                                                     | '1' – '128'     | 'Broadcast'     |
| # VersionNumber examples<br>InterfaceName.Update('VersionNumber', {'Module ID': '1'})<br>Value = InterfaceName.ReadStatus('VersionNumber', {'Module ID': '1'})<br>InterfaceName.SubscribeStatus('VersionNumber', None, FeedbackHandler)                         |                 |                 |

---

## Cable and Adapter Requirements

Please consult manufacturer for Cable and Adapter Requirements.

## Notes for the Device

---

### Serial communication

**Port Type:** RS-485

**Baud Rate:** 9600

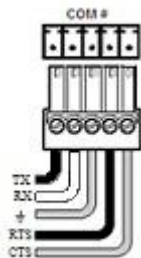
**Data Bits:** 8

**Parity:** None

**Stop Bits:** One

**Flow Control:** None

### Pin Assignments Diagram



Please consult manufacturer for Pin Assignments Diagram.

## Appendix A. Set Commands

|                                                    |                                       |
|----------------------------------------------------|---------------------------------------|
| Button Press Long Press Module ID 1 Button 1       | \xCA \xFE\xC1\x04\x01\x01\x00\x01\xAC |
| Button Press Long Press Module ID 1 Button 16      | \xCA \xFE\xC1\x04\x01\x10\x00\x01\xAC |
| Button Press Long Press Module ID 128 Button 1     | \xCA \xFE\xC1\x04\x01\x01\x00\x80\xAC |
| Button Press Long Press Module ID 128 Button 16    | \xCA \xFE\xC1\x04\x01\x10\x00\x80\xAC |
| Button Press Long Release Module ID 1 Button 1     | \xCA \xFE\xC1\x04\x02\x01\x00\x01\xAC |
| Button Press Long Release Module ID 1 Button 16    | \xCA \xFE\xC1\x04\x02\x10\x00\x01\xAC |
| Button Press Long Release Module ID 128 Button 1   | \xCA \xFE\xC1\x04\x02\x01\x00\x80\xAC |
| Button Press Long Release Module ID 128 Button 16  | \xCA \xFE\xC1\x04\x02\x10\x00\x80\xAC |
| Button Press Short Press Module ID 1 Button 1      | \xCA \xFE\xC1\x04\x03\x01\x00\x01\xAC |
| Button Press Short Press Module ID 1 Button 16     | \xCA \xFE\xC1\x04\x03\x10\x00\x01\xAC |
| Button Press Short Press Module ID 128 Button 1    | \xCA \xFE\xC1\x04\x03\x01\x00\x80\xAC |
| Button Press Short Press Module ID 128 Button 16   | \xCA \xFE\xC1\x04\x03\x10\x00\x80\xAC |
| Button Press Short Release Module ID 1 Button 1    | \xCA \xFE\xC1\x04\x04\x01\x00\x01\xAC |
| Button Press Short Release Module ID 1 Button 16   | \xCA \xFE\xC1\x04\x04\x10\x00\x01\xAC |
| Button Press Short Release Module ID 128 Button 1  | \xCA \xFE\xC1\x04\x04\x01\x00\x80\xAC |
| Button Press Short Release Module ID 128 Button 16 | \xCA \xFE\xC1\x04\x04\x10\x00\x80\xAC |
| Indicator Control Close Module ID 1 Button 1       | \xCA \x01(\x02\x01\x01\xAC            |
| Indicator Control Close Module ID 1 Button 16      | \xCA \x01(\x02\x10\x01\xAC            |
| Indicator Control Close Module ID 128 Button 1     | \xCA \x80(\x02\x01\x01\xAC            |
| Indicator Control Close Module ID 128 Button 16    | \xCA \x80(\x02\x10\x01\xAC            |
| Indicator Control Invert Module ID 1 Button 1      | \xCA \x01(\x02\x01\x02\xAC            |
| Indicator Control Invert Module ID 1 Button 16     | \xCA \x01(\x02\x10\x02\xAC            |
| Indicator Control Invert Module ID 128 Button 1    | \xCA \x80(\x02\x01\x02\xAC            |
| Indicator Control Invert Module ID 128 Button 16   | \xCA \x80(\x02\x10\x02\xAC            |
| Indicator Control Open Module ID 1 Button 1        | \xCA \x01(\x02\x01\x00\xAC            |
| Indicator Control Open Module ID 1 Button 16       | \xCA \x01(\x02\x10\x00\xAC            |
| Indicator Control Open Module ID 128 Button 1      | \xCA \x80(\x02\x01\x00\xAC            |
| Indicator Control Open Module ID 128 Button 16     | \xCA \x80(\x02\x10\x00\xAC            |
| Relay Control Close Module ID 1 Relay 1            | \xCA \x01\x18\x02\x01\x01\xAC         |
| Relay Control Close Module ID 1 Relay 4            | \xCA \x01\x18\x02\x04\x01\xAC         |
| Relay Control Close Module ID 128 Relay 1          | \xCA \x80\x18\x02\x01\x01\xAC         |
| Relay Control Close Module ID 128 Relay 4          | \xCA \x80\x18\x02\x04\x01\xAC         |
| Relay Control Invert Module ID 1 Relay 1           | \xCA \x01\x18\x02\x01\x02\xAC         |
| Relay Control Invert Module ID 1 Relay 4           | \xCA \x01\x18\x02\x04\x02\xAC         |
| Relay Control Invert Module ID 128 Relay 1         | \xCA \x80\x18\x02\x01\x02\xAC         |
| Relay Control Invert Module ID 128 Relay 4         | \xCA \x80\x18\x02\x04\x02\xAC         |

## Global Scripter Module Communication Sheet

Revision: 1/17/2019

---

|                                          |                               |
|------------------------------------------|-------------------------------|
| Relay Control Open Module ID 1 Relay 1   | \xCA \x01\x18\x02\x01\x00\xAC |
| Relay Control Open Module ID 1 Relay 4   | \xCA \x01\x18\x02\x04\x00\xAC |
| Relay Control Open Module ID 128 Relay 1 | \xCA \x80\x18\x02\x01\x00\xAC |
| Relay Control Open Module ID 128 Relay 4 | \xCA \x80\x18\x02\x04\x00\xAC |

---

## Appendix B. Update Commands

|                                     |                           |
|-------------------------------------|---------------------------|
| Relay Control Module ID 1 Relay 1   | \xCA \x01 \x01\x01\xAC    |
| Relay Control Module ID 1 Relay 4   | \xCA \x01 \x01\x01\xAC    |
| Relay Control Module ID 128 Relay 1 | \xCA \x80 \x01\x01\xAC    |
| Relay Control Module ID 128 Relay 4 | \xCA \x80 \x01\x01\xAC    |
| Version Number Module ID 1          | \xCA \x01\xBB\x01\x01\xAC |
| Version Number Module ID 128        | \xCA \x80\xBB\x01\x01\xAC |